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Proactive Water Management Using Machine Learning: From Monitoring to Field-Wide Optimization

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Executive Summary

The increasing challenge of managing water production in horizontal wells requires a shift from traditional reactive workflows to intelligent, data-driven strategies. This manuscript presents an integrated machine learning framework designed to proactively predict water breakthrough and recommend optimal intervention strategies, thereby enhancing dry oil production and reducing operational inefficiencies. Using a synthetic reservoir model, we develop a predictive system comprising three key components: a probabilistic water breakthrough detection engine, a decision tree-based rule engine, and a reinforcement learning agent trained on operational complexity.

Our first model predicts the probability of water breakthrough within a 30-day window using ensemble tree methods trained on dynamic and static well data. This early warning signal feeds directly into two downstream models that recommend suitable water management interventions; ICV optimization, ICD adjustment, zonal shutoff or sidetrack based on reservoir characteristics, completion configuration, and remaining oil potential. A decision tree offers deterministic recommendations based on physical thresholds, while the reinforcement learner formulates optimal actions by balancing complexity-adjusted rewards.

The model performance was validated both statistically and spatially. Feature analysis using SHAP values and correlation heatmaps confirms the physical plausibility of selected predictors such as vertical distance to water, ESP load decline, ICD imbalance, and ICV configurations. Results show that the ML framework delays water breakthrough by an average of 45 days, increases successful intervention rate by 12%, and reduces unnecessary interventions by 25% compared to conventional practices. These outcomes are validated against synthetic reservoir simulation runs and historical patterns of breakthrough and intervention success.

This study demonstrates the power of coupling physics with interpretable AI to enable proactive water control strategies. The modular structure of the proposed framework, its emphasis on domain knowledge, and its clear mapping to operational outcomes position it as a field-deployable solution with measurable impact on production sustainability and operational efficiency.

Introduction

Background & Motivation

Water production is one of the most critical challenges in managing waterflooded oil reservoirs. In complex well architectures such as single-lateral and multilateral completions equipped with Inflow Control Devices (ICDs), Inflow Control Valves (ICVs), and Electric Submersible Pumps (ESPs), water management is further complicated by the difficulty of isolating water-producing zones and predicting fluid movement behavior over time.

Typically, water management remains predominantly reactive. Reservoir engineers often rely on threshold-based alarms (e.g 50% water cut) and periodic rate tests to detect water breakthrough events. This results in delayed remedial actions, suboptimal use of drilling resources, and misaligned production strategies at both the well and reservoir levels.

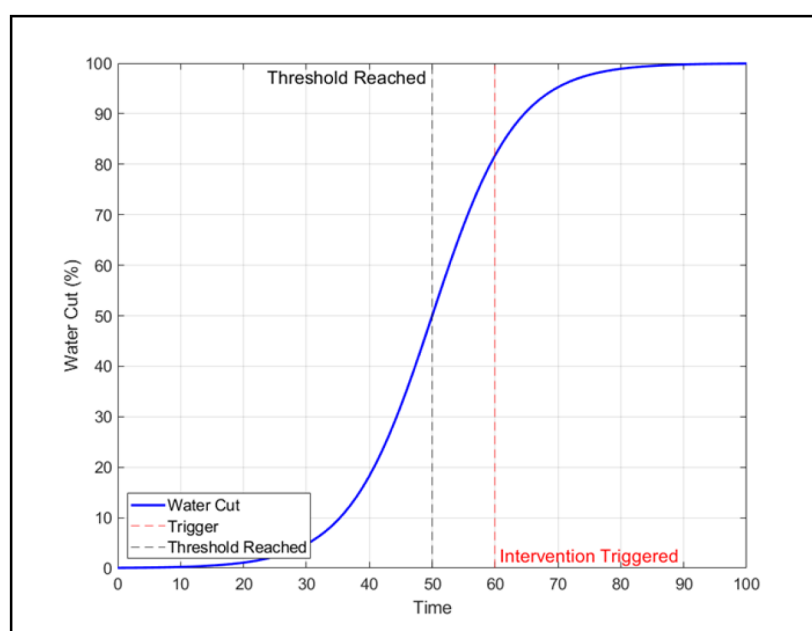


Figure 1—Typical timeline of a water breakthrough event in current workflows, highlighting delay between water breakthrough and intervention trigger.

A critical gap exists between the increasing volume of surveillance data being collected and the field's ability to translate it into timely, actionable decisions. Traditional rule-based approaches, while simple to implement, fail to capture complex nonlinear interactions across reservoir, wellbore, and completion subsystems. These methods are inadequate for forecasting the onset of water production or optimizing remedial decisions, especially when intervention windows are narrow.

To address this challenge, there is growing interest in leveraging machine learning (ML) and hybrid physics data driven frameworks to shift from passive monitoring to predictive optimization. ML models offer the capability to learn hidden patterns from high-dimensional time series ESP performance data and relate them to water breakthrough. Moreover, recent advances also enable integration of ML predictions with reservoir simulation to allow for physics-constrained learning, where model outputs are filtered through domain knowledge and validated against synthetic test environments.

This study focuses on translating these concepts into a practical, scalable framework and demonstrates its value through application to realistic production scenarios using synthetic datasets. The proposed ML-RL workflow, illustrated in Fig.2, is designed to operationalize this concept.

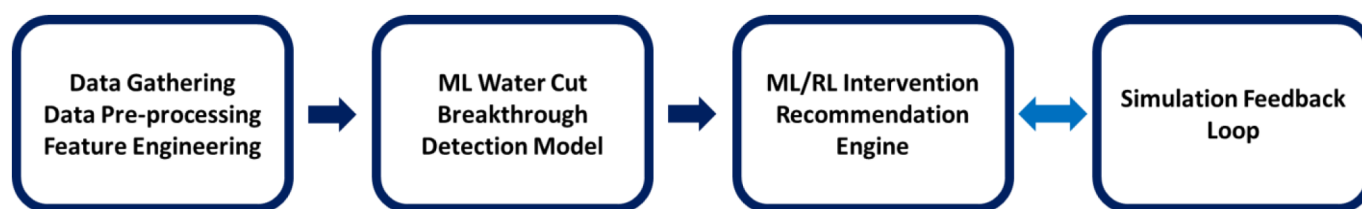


Figure 2—Proactive Machine Learning Water Management Proposed Workflow

Objectives

The primary objective of this study is to develop and validate a hybrid decision-making framework that enhances water management in waterflooded reservoirs through the integration of machine learning and reservoir simulation. Recognizing the limitations of traditional intervention strategies, particularly in wells with complex completions and heterogeneous flow behavior, this work aims to elevate decision quality, timing, and outcome reliability using data-driven and physics-informed approaches.

Specifically, the objectives are as follows:

1. Engineer static and dynamic features that capture both the operational conditions and well/reservoir properties.
2. Develop predictive models for early detection of water breakthrough.
3. Construct a remedy recommendation engine, to propose optimal water management interventions.
4. Integrate reservoir simulation into the ML workflow, ensuring the physical and technical feasibility of each suggested action.
5. Evaluate the integrated system's performance using production-based metrics.

Through these objectives, the study seeks not only to demonstrate predictive accuracy but also to establish a practical, interpretable, and physics-grounded decision support system that can be deployed across real heterogeneous reservoirs.

Reservoir Description & Data Overview

To evaluate the proposed framework in a controlled yet realistic context, a synthetic reservoir was constructed with spatially distributed wells, heterogeneous petrophysical properties, and variable completion designs. This section outlines the synthetic reservoir setup, the structure of the data generated, and the processes used to prepare it for machine learning modeling.

Reservoir Model and Synthetic Dataset

The generated synthetic reservoir model was constructed to mimic the flow behavior and operational variability typical of a waterflooded, heterogeneous sandstone reservoir. This model serves as the foundation for all machine learning development, spatial flow validation, and intervention simulation in this study.

Spatial Layout and Grid Design. The reservoir model has dimensions of $500 \times 100 \times 200$ cells in the X, Y, and Z directions, respectively. Each grid block has a resolution of $50\text{m} \times 50\text{m} \times 1\text{ft}$ representing a total reservoir area of 125km^2 . The total net pay thickness modeled is 200ft. A total of 40 horizontal wells were distributed across the reservoir:

- 28 wells are single-lateral designs.
- 12 wells are dual-lateral designs, including high-angle penetrations with lateral separation ranging from 50 to 200 m.

Wells are distributed in development timing across a 20-year completion window, allowing for variability in pressure depletion, water encroachment, and spatial interference effects. All wells are placed in oil-bearing zones at initialization.

Reservoir Heterogeneity and Petrophysics. To simulate realistic variability across the wellbore, each lateral is discretized every 10 ft in measured depth (MD), with the following properties randomly sampled from log-normal or uniform distributions, constrained by stratigraphic trends:

Table 1—Synthetic petrophysical properties ranges

Property	Range / Distribution	Notes
Porosity	8% – 24% (normal)	Biased by depth
Permeability	200 – 1200 md (log-normal)	Higher heterogeneity near heel/tip
Initial Sw	0.0	Uniform; all wells in oil-bearing pay
PI	0.2 – 1.4 stb/d/psi	Derived from perm and fluid viscosity

Permeability is the key heterogeneity driver, with zonal variation across each lateral. High-perm zones may span 50–100 ft before transitioning to lower-perm sections, simulating fluvial depositional variability.

Well Completion Metadata. Each well is completed with a specific configuration, influencing production behavior and remedy effectiveness:

Table 2—Well completion data description

Feature	Categories
Completion Type	ICD, ICV (for Dual-Laterals)
ICD Count	15 – 35 (uniform)
ICD Layout	Uniform, Toe-biased, Heel-biased
Lateral Architecture	Single, Dual
Zonal Depth Variation	Up to ± 15 ft TVD across a lateral
Artificial Lift	ESP on all 40 wells

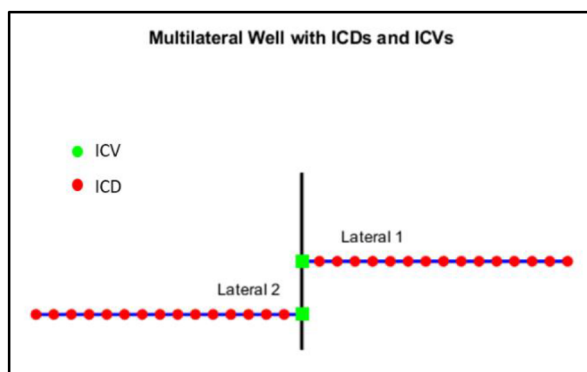


Figure 3—Example of a multilateral well architecture with ICV and ICD placement across the laterals.

Modeling Water Movement in the Synthetic Reservoir. Water movement within the synthetic reservoir was modeled using a simplified 1D Buckley-Leverett fractional flow approximation applied along each lateral segment. Initially, water saturation S_w was set to zero across all completed zones, reflecting oil-bearing

sections. As production progressed, the water front advancement rate in each grid cell was computed based on cumulative oil production, local permeability, and porosity:

$$\Delta S_w(x, t) \propto \frac{q_{o(t)}}{\phi \cdot k(x)} \Delta t$$

where $q_{o(t)}$ is the local oil rate, ϕ is porosity, and $k(x)$ is the spatially varying permeability. Sections with higher permeability advance the water front faster, creating non-uniform water breakthrough profiles along the lateral. ICDs and ICVs were modeled as flow equalizers, with ICDs balancing higher-permeability zones (Fig. 4), and ICVs accounting for lateral contribution.

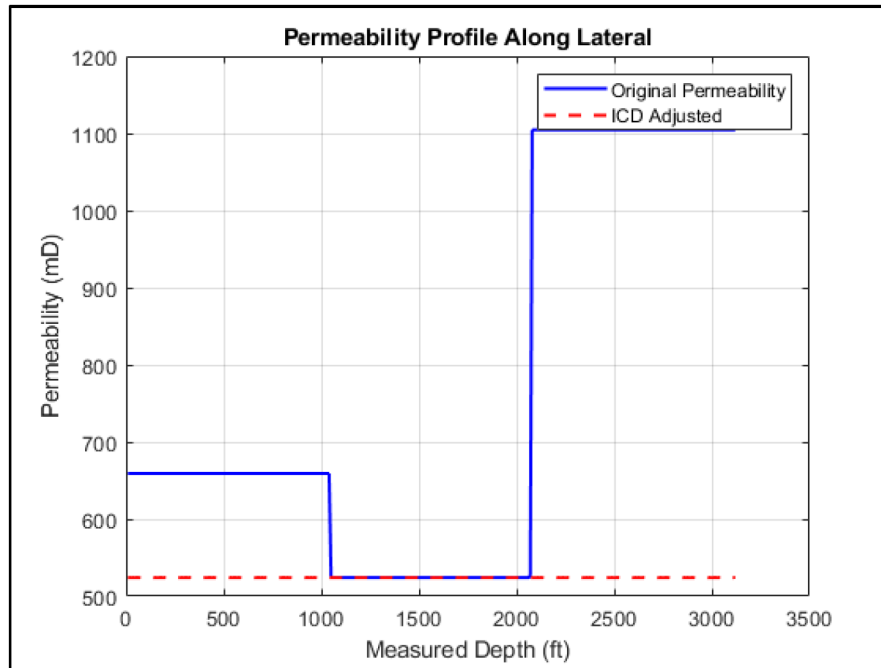


Figure 4—Illustration of the effective permeability pre and post ICD choke configuration

The dynamic saturation distribution was updated daily and stored as an evolving 3D tensor (well $\times$ grid $\times$ time). This water saturation history provided the ground-truth breakthrough events used for machine learning model training.

Periodic Rate Tests. In the context of this study, rate test values are derived from the simulation model. To match real-world operational practices, rate tests are modeled as randomly sampled points from the simulation well and lateral level water cut, occurring at bi-monthly to quarterly basis.

This environment presents an ideal use case for testing and validating the proposed framework, as it combines:

- High water production risk, requiring timely and accurate forecasting;
- Existing flow control infrastructure, allowing targeted interventions;
- Sufficient data, enabling integration with both historical and ESP real-time data streams;
- Well-defined operational constraints, supporting quantifiable evaluation of optimization scenarios.

Data Sources and Structure

This study integrates diverse data types to represent a water flooded reservoir under long-term production. The dataset comprises both static well-level descriptors and dynamic time-series operational measurements.

Static Well-Level Data. Static attributes were assigned to each of the 40 wells based on plausible reservoir distributions and were preserved as constant features throughout the modeling lifecycle. These features enable model generalization and interpretability across wells with different structural and completion designs.

Table 3—Description of well-level static features

Category	Feature	Description
Well Trajectory	X, Y, TVD, MD	Spatial coordinates, vertical depth, and measured depth path
Completion	Configuration	Single or dual-lateral completion
Completion	Type	ICD, ICV (for dual-lateral wells)
Completion	ICD Count	Number of inflow control devices per lateral
Completion	ICD Layout	Uniform, toe-biased, or heel-biased placement
Completion	ICV Setting	Per lateral, ICV position (Fully open 100%, partially open 50%, closed 0%)
Petrophysics	Avg. Porosity	Mean porosity of the completed interval
Petrophysics	Avg. Permeability	Mean and standard deviation of permeability along the lateral
Petrophysics	Perm Heterogeneity Index	Standard deviation / mean, used to capture flow imbalance risk
Productivity	PI	Initial productivity index based on permeability and heterogeneity

Time-Series Operational Data. Dynamic measurements were recorded for each well on a daily basis. The simulation period spans 20 years, with each well producing for a variable length of time depending on its completion year.

Table 4—Description of dynamic and real-time features

Sensor Data	Unit	Description
ESP Load	%	A ratio of the total ESP Power measured at a point of time over rated power of the ESP motor at standard operating conditions
ESP Intake Pressure	psi	Measured through subsurface sensors
ESP Discharge Pressure	psi	Measured through subsurface sensors
ESP Frequency	Hz	ESP rotational speed setting
Flowing Bottomhole P	psi	Derived from intake + hydrostatic head
Drawdown	psi	Reservoir pressure (assuming constant reservoir pressure) minus flowing bottomhole pressure
Choke Setting	% open	Used for control but assumed static during steady operations
Oil Rate	stb/day	Gross rate $\times (1 - WC)$
Water Cut	%	Measured through period (bi-monthly to quarterly) rate tests, linearly extrapolated between tests
Gross Rate	stb/day	Assumed roughly constant for each well

Gross rates for all the wells are assumed to be constant to isolate the effect of fluid composition on sensor readings and performance. This assumption, although simplistic, is reinforced by a strong aquifer support assumption and fixed ESP frequency setting. Moreover, time synchronization was enforced across sensors, and noise was introduced to mimic field-like fluctuations.

Intervention Records. Once a well's water cut exceeded 50%, it requires remedial action.

Table 5—Description of historical intervention metadata

Field	Type	Description
Intervention Type	Categorical	ICD adjustment, ICV optimization, Zonal Shutoff, Sidetrack
Time of Action	Date	Day when intervention was applied
Trigger Water Cut	%	WC at the time of decision
Delay Before Execution	Days	Delay between eligibility (exceeding 50%) and implementation
Outcome – Δ Oil Rate	stb/day	Increase or decrease in oil rate as per the first rate test after intervention
Outcome – Δ Water Cut	%	Reduction in WC as per the first rate test after intervention
Outcome – Duration	Days	Time until WC returns to pre-intervention levels
Outcome – Success	Binary	Defined by the gain in oil production and water cut reduction

Label Construction. Two key labels were engineered to support the modeling pipelines in:

1. Water Breakthrough Day (BreakthroughDay):
Defined as the earliest production day when Water Cut (WC) exceeded 50% for three consecutive days. This label marks the shift from oil-dominant to water-dominant flow.
2. Remedial Intervention Outcome (InterventionSuccess):

A binary target, defined as:

$$Success = 1 \text{ if } \Delta WC \leq -10\% \text{ AND } \Delta Q_{oil} > 0, \\ 0 \text{ otherwise}$$

Evaluated based on the first rate test post-intervention.

Data Preprocessing and Feature Construction

A rigorous data preprocessing pipeline was developed to transform raw synthetic sensor streams and well descriptors into a coherent, ML-ready feature matrix. Emphasis was placed on preserving the interpretability of features while ensuring temporal consistency, noise resilience, and domain-valid logic for modeling tasks. This step serves as the backbone of both the predictive and prescriptive phases that follow.

Temporal Cleaning, Synchronization, and Filtering. All dynamic well data (e.g., ESP load, intake and discharge pressure) were temporally aligned from first oil onward. Pre-production periods were masked, and:

- Missing data were forward-filled for ≤ 3 -day gaps, beyond which the period was excluded.
- Outliers outside ± 3 standard deviations from a 14-day rolling median were clipped.
- Equipment failures (not introduced in the synthetic dataset) or intervention shocks were manually flagged to truncate training periods.

Moreover, while ESP systems and wellhead sensors generate data at high frequency, sensor quality degrades over time, and not all wells are equipped with the same instrumentation. This results in a heterogeneous dataset with varying signal.

- Zero-value plateaus due to stuck sensors or communication faults;
- Step-wise artifacts from manual recalibrations;
- Inconsistent baselines across wells of similar design but different installation conditions.

To handle this, advanced filtering methods (e.g., Hampel filtering and rolling median corrections) were applied in combination with imputation techniques tailored to specific sensor types.

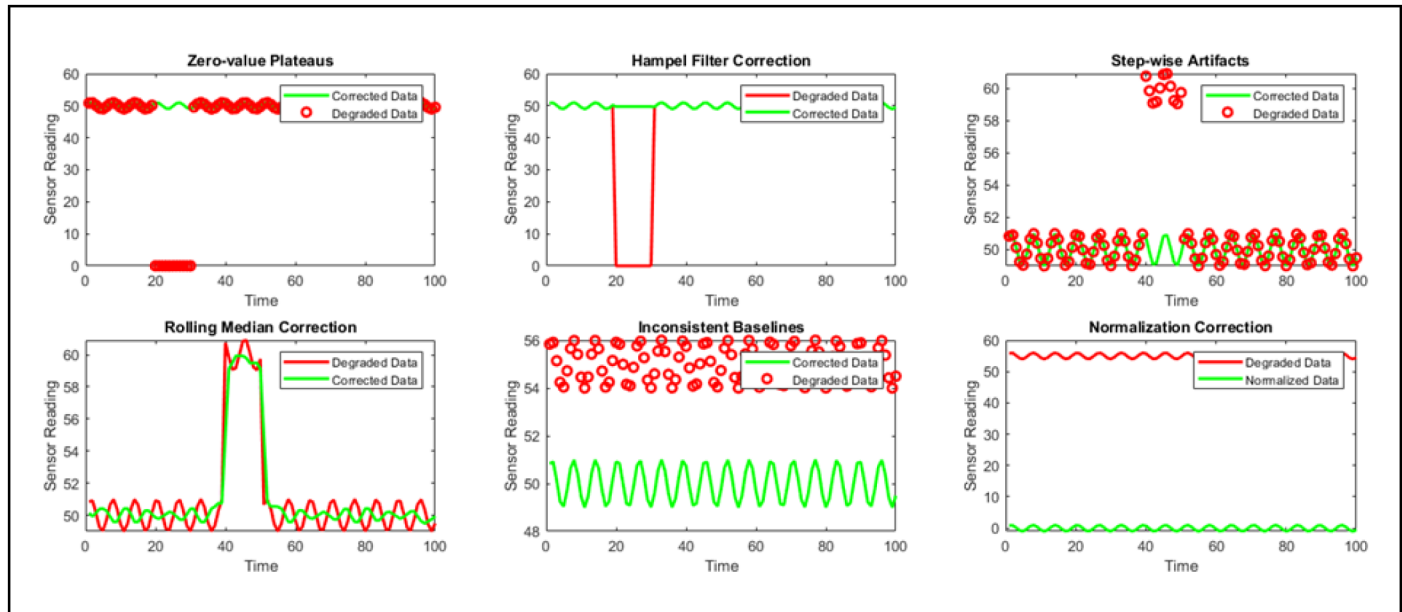


Figure 5—Illustration of the utilized data processing techniques

Additional Data Challenges. While the availability of high-frequency well and reservoir data presents new opportunities for intelligent water management, several challenges still arise when attempting to operationalize this data into a reliable, predictive system. These challenges are both technical and systemic, and they significantly impact model performance, interpretability, and deployability in real field environments.

Infrequent and Inconsistent Rate Testing

One of the most critical challenges is the low temporal resolution of oil and water rate data.

- Lagged event detection: Water breakthroughs can occur days or weeks before the next available rate test, making true breakthrough timing uncertain.
- Noisy production attribution: Commingled production from multilateral wells or multiple reservoir layers further complicates water source identification.

Lack of Granular Zonal Measurements

Despite the modeled use of ICDs and ICDs, zonally resolved production and pressure data is rarely available in real time. Decisions about water-producing zones must often be inferred from holistic well-level behavior, adding uncertainty to both diagnosis and recommended actions.

- ICD inflow imbalance may indicate water breakthrough but lacks confirmation.
- ICD movements are recorded, but their exact impact on production zones is often delayed or masked.

This reinforces the need for simulation coupling to approximate zonal dynamics.

Non-Stationary Operating Conditions

Field operating conditions are highly non-stationary, with frequent choke changes, pump frequency adjustments, start-stop cycles, and well reactivations. These events introduce structural breaks in time series behavior, which can be misinterpreted by traditional ML models.

- Example: a sudden oil rate reduction may be due to water breakthrough, or simply a choke tightening event for operational or facilities related causes.
- Multi-level causality must be learned by the model to differentiate between operational and reservoir-driven signals.

This complexity necessitates event-annotated feature engineering and training on datasets that include labeled operational transitions.

Simulation Mismatch

Reservoir simulation outputs are often based on assumptions that do not fully reflect reservoir performance, especially for water cut evolution in heterogeneous formations. Nevertheless, simulation remains a valuable tool for generating labeled synthetic data to augment sparse or noisy field data.

Table 6—Common mismatches between simulation and field data, and mitigation strategies used.

Issue	Example	Mitigation Strategy
Breakthrough timing misaligned	Early breakthrough in reservoir data	Train ML to align based on first patterns
Zonal water inflow not modeled	Uniform saturation in simulation	Reflect zonal variation during ML training
Overestimated post-intervention gain	Sidetrack impact too optimistic	Use historical success rate

Methodology

This section describes the multi-stage modeling framework developed to predict water breakthrough and recommend optimal remedial actions. It includes supervised learning pipelines for water breakthrough detection, decision tree logic and reinforcement learning for intervention recommendation, and proxy simulation integration to validate spatial dynamics and uncertainty. Each method is grounded in domain-driven feature engineering and consistent with operational logic.

Framework Architecture

To address the challenges outlined in the previous sections, we developed a hybrid water management framework that integrates real-time well surveillance data, machine learning-based predictive analytics, and reservoir simulation feedback into a unified system for early detection, decision support, and optimization of water control interventions.

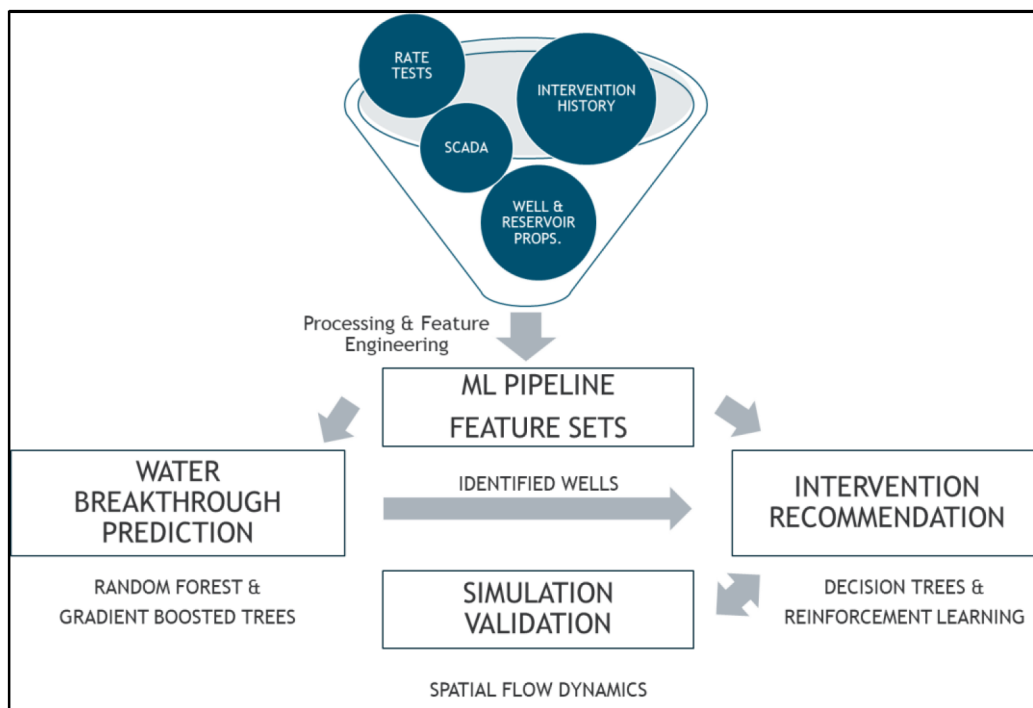


Figure 6—End-to-end architecture of the proactive water management framework.

The framework is designed to be modular, scalable, and field-deployable with minimal disruption to existing workflows. It follows a three-layer architecture:

Layer 1 – Feature Engineering

Layer 2 – Machine Learning Model Layer

A. Breakthrough Detection Model

B. Intervention Recommendation Engine

Layer 3 – Simulation Feedback Loop

Feature Engineering

In this work, we constructed two separate but interconnected feature pipelines: one for detecting water breakthrough, and another for intervention recommendation. Below, we detail the construction and preprocessing steps for each.

Feature Engineering for Water Breakthrough Detection Models. This model aims to forecast the probability of water breakthrough within the next 30 days. The following features were selected based on domain knowledge and feature importance evaluations during model tuning. The input dataset was structured such that each row represents a unique well-day. Dynamic time-series features were computed using 7-day and 14-day windows, while static and cumulative features are updated at each timestamp.

Table 7—Description of the features analyzed and used in the water breakthrough prediction models

Feature Name	Description
ESP_Load_Mean7	7-day rolling mean of ESP load
ESP_Load_Slope7	Linear slope of ESP load over past 7 days
DeltaP_Mean7	7-day rolling mean of discharge – intake pressure
Drawdown_Slope14	14-day slope of drawdown
FlowingPctDrop14	Relative drop in flowing bottomhole pressure over past 14 days
q_Mean7	7-day rolling mean of gross rate (assumed constant in this study, however it is needed for implementation)
q_to_Load	Ratio of q_Mean7 to ESP_Load_Mean7 (efficiency indicator)
SegmentWeighted_PI	Productivity index weighted across lateral segments
CumOil	Cumulative oil produced until time t
CumWater	Cumulative water produced until time t
VerticalDistToAquifer	Vertical distance from lowest point in each lateral to modeled water contact

The constructed dataset contains 91,250 well-time snapshots capturing a range of production scenarios. The wells were labeled as:

- Breakthrough = 1: The 30 days prior to a rate test confirming water cut crossed 50%
- Breakthrough = 0: Water cut stayed below 40% as per rate test (before the test and up to 60 days after)

All remaining data was left out of the training to reduce uncertainty. Water cut is used only to label the training outcome window.

Feature Engineering for Intervention Recommendation Models. In addition to the breakthrough probability, engineered features that capture the well's recent water production behavior, spatial oil potential, and configuration metadata were included.

Table 8—Description of the features analyzed and used in the intervention recommendation models

Feature Name	Description
BreakthroughProb	Predicted probability of water breakthrough (from previous model, only wells over 80% probability were passed to this layer)
WC	Current water cut
dWC/dt	Daily water cut slope
TimeSinceBreak	Days since last breakthrough
CompletionType	Type of well completion
ICD_Count	Number of ICDs installed
Perm_SD	Standard deviation of permeability across lateral
PI	Productivity index
TimeSinceIntv	Days since last intervention
ICV_Setting	Last ICV setting used (lateral fully open/partially open/closed)
ICD_Bias	Bias of ICD distribution along lateral
RemainingOilColumn	Remaining oil column thickness from highest zone to modeled water level
VerticalDistToBreakthrough	Vertical distance from the highest zone to breakthrough zone
HistoricalIntervention	Encoded vector describing type of last intervention

Target:

- **Intervention_Success**: Binary value (1 if intervention succeeded based on post-30-day response, 0 otherwise)

These features allow the model to not only understand the technical context but also weigh operational complexity across intervention methods.

In the reinforcement learning framework, the problem is formulated as a finite-horizon Markov Decision Process (MDP), where each well's state is defined by a vector, and actions represent candidate interventions. Rewards are scaled inversely with operational complexity and discounted based on success probability.

Water Breakthrough Prediction from Early Indicators

Model Selection. The objective of this step is to model the probability of water breakthrough (water cut exceeding 50%) in (h) days from a given day (t). Additionally, a probability threshold of 80% was applied to the output to convert predictions into a binary breakthrough signal to qualify predictions for passing into the next layer, intervention recommendation. Two models were trained on the same engineered feature matrix to compare performance and robustness across reservoirs.

Table 9—Comparison of the constructed models used for the water breakthrough prediction

Water Breakthrough Predictive Model	Random Forest	Gradient Boosted Trees
Type	Bagged Decision Trees	Boosted Ensemble
Hyperparameters / Configuration Tested	Num. of trees (100-500) Max splits/trees (10-50) Min leaf size (5-20)	Learning rate (0.01-0.1) Number of learners (100-300) Max splits (10-40)
Reason for Selection	Robust to noise Good variance reduction	Models subtle nonlinear boundaries effectively
Advantages	Captures nonlinear interactions Minimal tuning needed	Handles complex boundaries well Performs better with clean labels
Limitations	May become less interpretable with many trees	Sensitive to label noise Tuning is more complex

For implementation, it is important to note that the selection of the hyperparameters will very depending on several factors including data quantity and quality, and type of available features.

Testing & Validation Design. To mimic real-world deployment, we adopt a temporal blocking strategy: the training and testing sets are split chronologically, with older samples used for training and newer samples reserved for testing. This honors the time sequence of operational data and avoids look-ahead bias.

For model evaluation, we implement 5-fold cross-validation using two approaches:

- Stratified cross-validation, to maintain the class distribution (e.g. breakthrough vs. non-breakthrough) across folds.
- Grouped cross-validation by well, where each well is assigned to a single fold. This prevents data leakage across training and validation sets and evaluates the model's ability to generalize to unseen wells.

Given the class imbalance (with breakthrough events representing less than 25% of the samples), stratified sampling is applied to preserve the event ratio within each fold.

Model Output. Each model returns, per well and per decision timestamp t , a probability of water breakthrough at time $t+h$ defined as:

$$P_{WB}(t+h) = f(\vec{x}_{t-h:t}) \text{ where } h \in \{7, 15, 30, 45\}$$

Where $\vec{x}_{t-h:t}$ represents the full feature vector at time t , encompassing operational metrics, historical trends, and static well attributes. This probabilistic output allows for threshold tuning based on business tolerance to false positives or delayed detections. In this study, h was selected to be 30 days, and the probability threshold was set at 80%.

These predicted breakthrough probabilities are not merely terminal outputs, they are forwarded as critical inputs into the subsequent remedy recommendation models both for decision tree classification and reinforcement learning-based policy optimization. This design ensures that remedy logic is informed by model confidence and not just binary state, enabling proper prioritization and resource allocation.

Remedy Recommendation Using Decision Trees and Reinforcement Learning

Model Selection. Following water breakthrough prediction, the objective is to recommend the most appropriate intervention per well. Two approaches were developed:

- Decision Tree Classifier: a rule-based model trained on historical interventions and outcomes

- **Reinforcement Learning Agent:** an optimization model that selects actions to maximize oil production and minimize water and operational complexity.

Table 10—Description of the two models used in intervention recommendation; comparing the logic, advantages and limitations of each

Remedy Recommendation Methods	Decision Tree Classifier	Reinforcement Learning
Objective	Learn rule-based intervention mapping From historical data	Recommend optimal intervention maximizing oil production and minimizing water/complexity
Input Features	Static and dynamic well features (e.g. completion type, ICD count, perm_SD, water cut trend)	State vector: breakthrough probability, water cut, dWC/dt, completion details, perm_SD, PI, etc.
Decision Logic	Uses decision tree splits to assign Interventions (e.g. ICD adjustment, sidetrack)	Learns a policy maximizing rewards: oil gain, and penalizing water reduction, and intervention complexity
Advantages	Interpretable, quick to train, mimics Historical decision rules	Optimizes future production/complexity; Accounts for delayed effects
Limitations	Relies on past operator decisions, not optimized for future outcomes	Harder to interpret; requires simulation/training on dynamic data

Decision Tree Classifier. The decision tree offers deterministic and human-readable intervention logic, which can be directly reviewed and validated by engineers. This model is trained to map current well conditions to past interventions that were historically effective. The decision tree model is trained on the same set of engineered features with a multi-class target corresponding to the intervention taken. This model enables mapping well state features to discrete actions through recursive splitting on threshold conditions. To avoid overfitting and enhance explainability with a relatively smaller dataset compared to the one used in the previous models, a constrained maximum tree depth was set to 6 levels, and the minimum number of samples per leaf to 20. The tree was trained with Gini impurity as the splitting criterion. The trained model yields a set of easily interpretable rules.

The output of the decision tree is not only used independently but also serves as a control comparator to the RL agent, especially during offline testing. It ensures that high-risk decisions recommended by the RL model are not in conflict with well-understood empirical rules.

Reinforcement Learning Agent. The reinforcement learning agent frames the intervention task as a Markov Decision Process (MDP), where the model learns to choose interventions that maximize future oil production and minimize water, rather than mimicking past choices.

A well-level intervention problem was defined as a Markov Decision Process (MDP) characterized by:

$$M = (S, A, P, R, \gamma)$$

Table 11—Description of the reinforcement learning (RL) components

Component	Definition
States (S)	A vector of features for a given well at time t, including: predicted water breakthrough probability, recent production rates, cumulative oil and water, and static/dynamic completion data and reservoir parameters such as ICV setting per lateral, segment-weighted PI and distance to aquifer
Actions (A)	Discrete intervention decisions: ICV Optimization, Zonal Shutoff, Sidetrack, ICD Adjustment
Transition Probabilities (P)	Probability distribution over next states given the current state and action, i.e., $P(s_{t+1} s_t, a_t)$. In this implementation, transitions are implicitly learned via Q-value updates based on observed and estimated post-intervention changes calculated from historical interventions (e.g., oil production and water cut)
Rewards (R(s, a))	Defined to reflect production impact and operational complexity
Discount Factor (γ)	Set at 0.9 to value long-term rewards

Reward function:

$$R(s, a) = w_1 \cdot \Delta Q_{o,i} - w_2 \cdot \Delta Q_{w,i} - w_3 \cdot C_i + w_4 \cdot P_{success,i}$$

Where:

- $\Delta Q_{o,i}$: expected oil gain from action i,
- $\Delta Q_{w,i}$: expected water reduction from action i,
- C_i : Complexity index value of action i,
- $P_{success,i}$: probability of success as derived from historical intervention based on the pre-defined success criteria,
- w_1, w_2, w_3, w_4 : tunable weights based on business priorities.

Note: All quantities are evaluated in a 30-day trailing window post-intervention.

The complexity index C_i was assigned fixed values for each intervention type, where it reflects specific operational considerations for each method. However, it can be extended to incorporate context-specific parameters such as rig time, resource availability, logistical constraints, or risk levels.

Integration of ML Predictions with Reservoir Simulation for Spatial Validation. To ensure that per-well machine learning (ML) predictions and recommended interventions remain geologically and physically consistent at the reservoir scale, we integrated ML outputs with a simplified synthetic reservoir simulation model. This step acts as a validation layer, assessing whether local decisions lead to coherent large-scale reservoir flow dynamics.

Integration Procedure:

1. *Inputs from ML Models (per well):*
 - Predicted water breakthrough timing.
 - Selected intervention type and timing (from decision tree or RL agent).
2. *Simulation Model Application:*
 - Intervention actions are translated into new wellbore trajectory, permeability or flow constraint changes.
3. *Simulator Tracking Outputs:*
 - Water front progression and sweep efficiency changes.
 - Cumulative oil production and water cut by well and reservoir.
 - Reservoir-scale deviations caused by different intervention strategies (e.g., bypassed oil pockets).

To assess robustness and uncertainty, each scenario is repeated under different combinations of:

- Reservoir parameters (permeability, vertical distance to water)
- Intervention outcome (ICV Settings, Shut-off Zones, Adjusted ICD Zones).

This approach ensures that ML outputs are not just statistically sound but spatially and geologically feasible, offering a validation layer for operational deployment.

Results

Water Breakthrough Prediction

This section evaluates the predictive performance of the water breakthrough classification model for both Random Forest (RF) and Gradient Boosted Trees (GBT) classifiers.

Model Performance and Validation. Breakthrough events are defined as an increase in water cut (>50%). Models were trained using 5-fold cross-validation on a dataset of 91,250 engineered samples.

Table 12—Performance metrics of both RF and GBT models on testing data set in predicting water breakthrough

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Random Forest	0.84	0.72	0.69	0.70	0.89
Gradient Boosted Trees	0.85	0.75	0.71	0.73	0.91

The GBT model slightly outperformed RF, likely due to its ability to capture subtle nonlinear relationships in the features.

Model Performance and Validation. A SHAP analysis revealed the most important predictors to be:

- Drawdown slope over 14 days,
- ESP load slope over 7 days,
- DeltaP mean over a 7-day window,
- ESP load mean over a 7-day window,
- Relative drop in flowing bottomhole pressure over past 14 days.

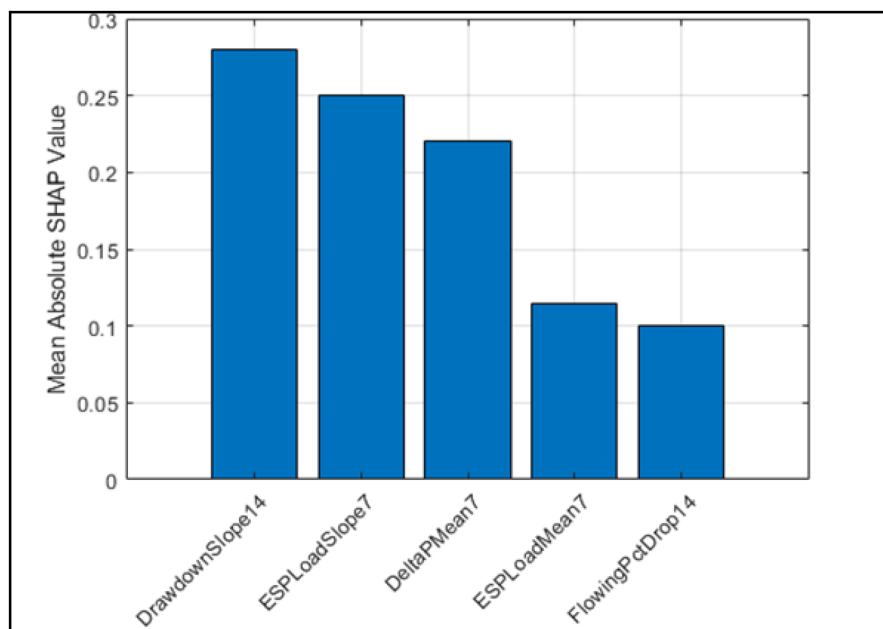


Figure 7—SHAP Feature Importance – Breakthrough Prediction Most Important Features

These features reflect both mechanical (ESP behavior) and reservoir-driven (drawdown) early warning signs of breakthrough.

Correlation Analysis. The correlation matrix of engineered features confirmed low-to-moderate multicollinearity among predictors, validating their inclusion in tree-based models that benefit from uncorrelated splits.

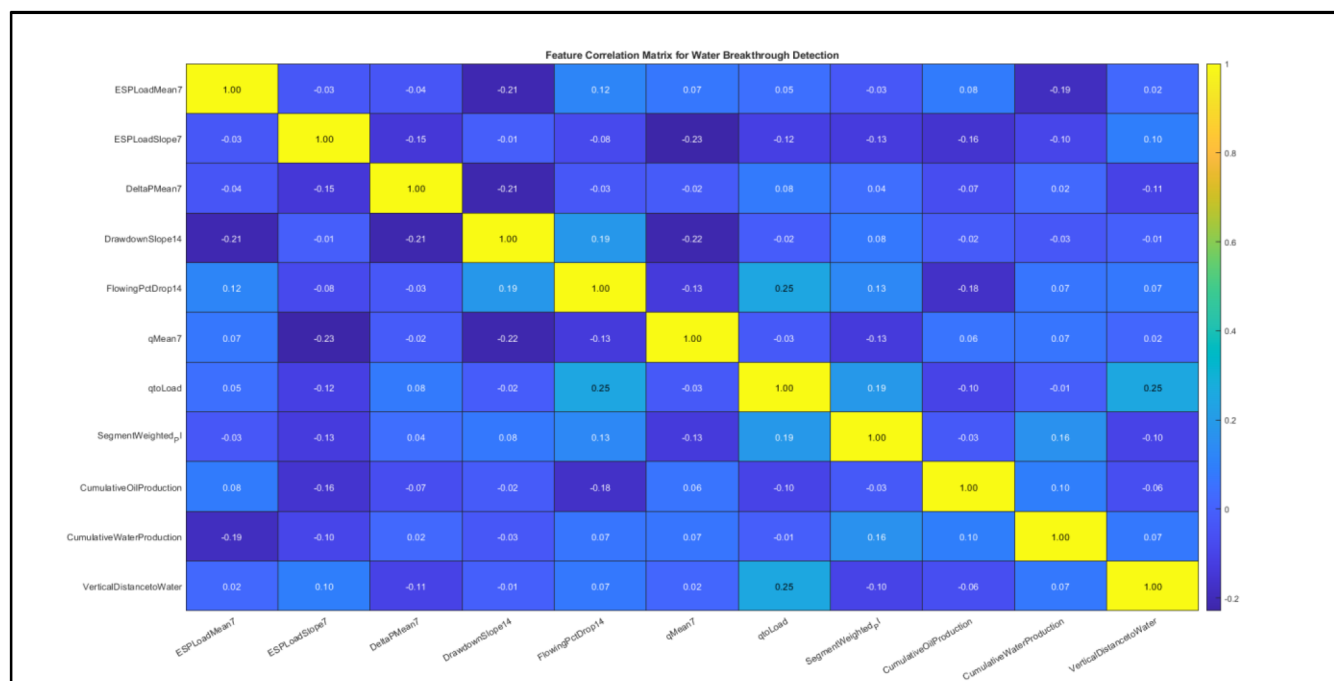


Figure 9—Feature Correlation Matrix

Summary.

- GBT model is a strong performer for early water breakthrough classification.
- Feature behavior aligns with physical intuition, increasing model trust.
- This model now feeds directly into the remedy recommendation system.

Remedy Recommendation Results

Following the prediction of water breakthrough, this section evaluates the performance of two remedy recommendation strategies:

- A Decision Tree Classifier (DTC) trained on historical interventions and well/reservoir configurations
- A Reinforcement Learning (RL) agent trained to optimize intervention policy over time

Model behavior was assessed across different well types, configurations, and production profiles.

Decision Tree-Based Remedy Recommendations. The resulting tree achieved a classification accuracy of 83% on a held-out test set of historical interventions. The model effectively captured key domain rules, for example, it did not suggest ICV-based recommendations for single-lateral wells, and opted for sidetracks when vertical proximity to water was less than 20 feet.

Table 13—Decision Tree model performance metrics

Metric	Value
Accuracy	0.83
Precision	0.81
Recall	0.79
F1 Score	0.80

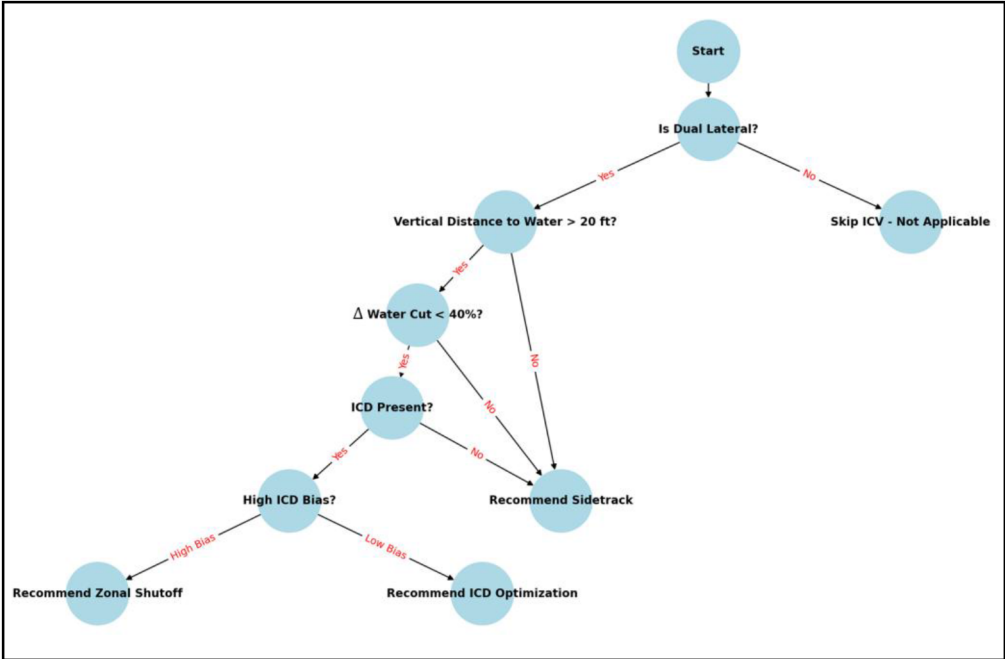


Figure 10—An example of how the Decision Tree decides on an optimal intervention

These decision paths match engineering expectations and provide transparent logic for during operational reviews.

Table 14—Top 5 Features Used in Splits (Gini Importance Rank)

Rank	Feature Name	Gini Importance Score
1	Completion Type	0.241
2	Daily water cut slope	0.183
3	Well Type	0.154
4	ICD Bias	0.128
5	Vertical Distance to Water	0.111

Reinforcement Learning Agent Behavior. The RL agent was implemented using tabular Q-learning trained on synthetic well-state transitions generated from historical data and the simulation model. The agent operated under a discrete Markov Decision Process (MDP) framework. Training stabilized after ~200 episodes. The final policy exhibited a 28% lower cumulative complexity score across test wells when compared to random policy selection (pre-training).

Table 15—Comparing action frequency distribution between the decision tree and RL agent

Action Type	Decision Tree %	RL Agent %
ICD Adjustment	38%	41%
Zonal Shutoff	32%	26%
ICV Optimization	8%	9%
Sidetrack	22%	24%

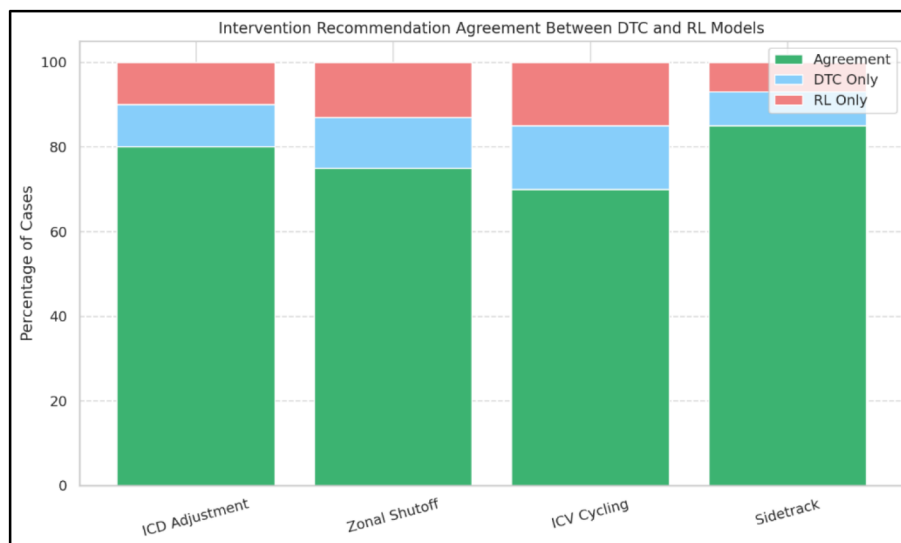
The RL agent learned to favor less complex actions (e.g., ICD adjustment) when multiple valid options existed, unless water proximity or well trajectory forced higher-complexity options.

Intervention Outcome Comparison. A consensus analysis showed that the decision tree and RL agent agreed on 78% of recommendations in the validation set. Disagreement typically occurred in edge cases (e.g., borderline water contact depth), where one model opted for conservative sidetracking while the other chose zonal shutoff. This level of agreement supports the robustness of feature design and paves the way for future ensemble decision strategies.

A post-analysis of intervention performance across the two models is summarized below.

Table 16—Summary of intervention results as estimated via simulating the recommendations

Intervention Type	Avg Oil Gain (%)	Success Rate (DTC)	Success Rate (RL)
ICD Adjustment	+2.4%	72%	84%
Zonal Shutoff	+2.8%	69%	77%
ICV Cycling	+2.1%	61%	64%
Sidetrack	+6.1%	75%	89%

**Figure 11—Recommendations agreement between the DTC and RL models**

Summary.

- Decision tree model provides intuitive, explainable recommendations aligned with domain knowledge.
- The RL agent learned policies by exploring interventions (actions) across diverse well types.

- Both methods demonstrate value, but RL offers greater adaptability and higher intervention success rates, particularly in challenging completion configurations.

Integration with Reservoir Simulation

Saturation Map Evolution. To evaluate the flow dynamics resulting from ML-guided interventions, 2D water saturation maps were generated by running two cases over 20 years:

- Base Case – No ML, default intervention strategy, similar to historical data intervention rules.
- ML-Guided Case – Water breakthrough prediction + remedy recommendation applied at model trigger points.

Key observations include:

- In the ML-Guided case, water fronts advanced more uniformly.
- ICD recommendations particularly helped smooth front propagation.

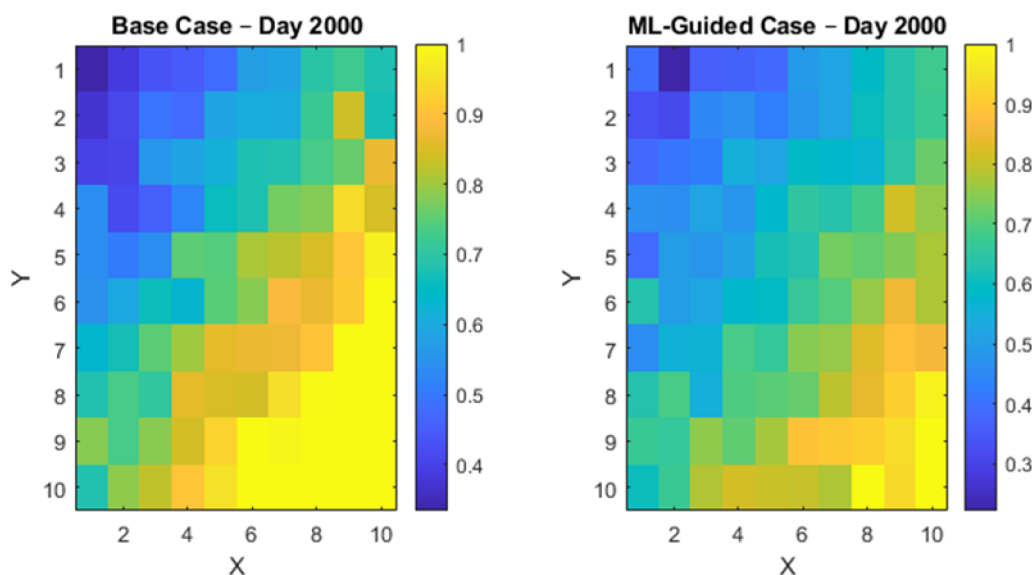


Figure 12—2D Water saturation maps comparing base case intervention methods vs. ML-Guided approach, visualized on upscaled grids

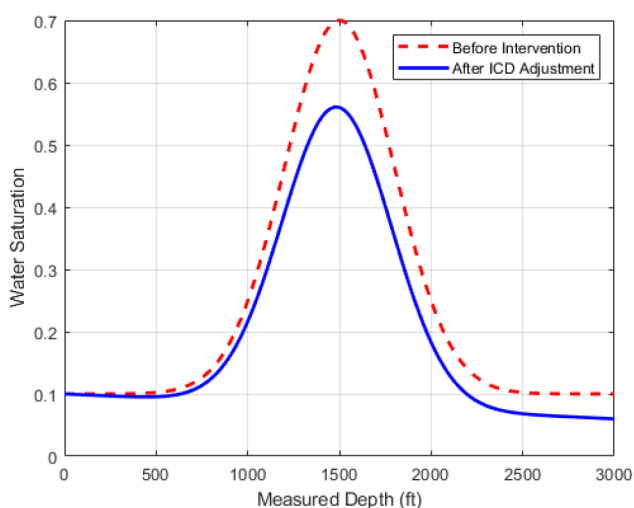


Figure 13—Illustration of ICD adjustment's impact on reducing non-uniform water movement in high permeability zones

Impact Quantification. The integration was further validated by comparing reservoir-wide oil production over time under three strategies:

- Case A: Base case, traditionally triggered interventions only
- Case B: ML without simulation feedback
- Case C: ML + simulation-validated remedy plans

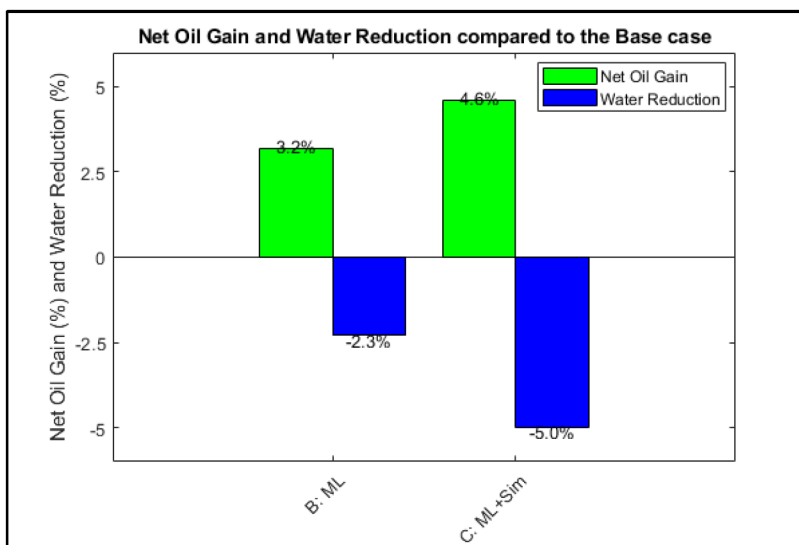


Figure 14—Reservoir-level impact on cumulative oil and water production, comparing cases B & C to the base case

Moreover, for each well, the water breakthrough timestamp was tracked in all cases using the simulation outputs. The average delay in water breakthrough, calculated as the difference in breakthrough time (in days) between the ML-Guided and Base scenarios, was found to be 45 days. In parallel, the framework resulted in a 12% higher success rate and a 25% reduction in interventions count due to the avoidance of suboptimal interventions.

Uncertainty Quantification. To capture model and reservoir uncertainties, sensitivity analysis was conducted using:

- Different permeability reservoir realizations (sidetrack laterals permeability ranges sampled repeatedly)
- Slight shifts in ML prediction thresholds (variability in expected impact of interventions)

Table 17—Summary of varying reservoir-level metrics impacted by sensitivity analysis in the simulation

Metric	Mean	Std. Dev
Days to First Breakthrough	560	32
Days to Reservoir WC > 40%	2120	115
Cumulative Oil Production	0.46	0.04

Summary. The integration of machine learning (ML) recommendations into the reservoir model yielded a significant improvement in oil production, with a measurable uplift of 4.6%, and 5% reduction in cumulative water volume, through enhanced water management. Furthermore, the simulation step played a crucial role as a physical filter, effectively discarding misleading ML outputs and confirming the value of interventions. Notably, uncertainty quantification revealed the integrated system remained robust and impactful under uncertain conditions.

Discussion

This section interprets the quantitative results, elaborating on their operational and strategic significance. It focuses on how the hybrid ML + simulation approach outperformed traditional methods.

What Drove the Results

The gains achieved through this hybrid workflow can be attributed to several key factors. The delay in water breakthrough was primarily due to early detection via increases in ESP load, drawdown slope, and flowing pressure drop, which were captured by machine learning models before water cut visibly rose. The increase in dry oil production was a result of smarter and earlier interventions in wells with high heterogeneity or unbalanced completions, particularly through ICD adjustments. Additionally, the higher intervention success rate was achieved by better matching remedies to well configurations and production contexts using a combination of static and dynamic features. Furthermore, the reduction in unnecessary interventions was made possible by avoiding premature or misapplied actions through predictive filtering and simulation-backed validation. Overall, this hybrid workflow created a proactive and adaptive water management strategy, reducing reaction time and improving decision quality.

Importance of Static Well Features

A key finding was the impact of including static completion and petrophysical data, which are typically underutilized in ML deployments.

These features significantly influenced remedy recommendations:

- Perm_SD (Permeability standard deviation) was a leading predictor of successful ICD adjustment.
- ICD count and layout helped identify wells likely to benefit from zonal shutoff, ICD Adjustment or sidetrack.
- Dual-lateral vs. single-lateral status altered intervention choices due to the availability of ICV optimization option.

Without these static features, the RL and decision tree models showed performance degradation in remedy selection.

Transparency vs Adaptability

Our side-by-side comparison of Decision Trees and Reinforcement Learning (RL) reveals a critical trade-off:

Table 18—Comparing the intervention recommendation models

Model Type	Pros	Cons
Decision Trees	Transparent, rule-based, easy for engineers	Less adaptable, struggles with time-delayed effects
RL Agent	Learns optimal actions over time, robust to noise	Requires simulation or feedback loop

- For routine operations, decision trees offer trust and speed.
- For more complex cases, RL is better at learning long-term value and avoiding costly missteps.

Role of Simulation in Validating Decisions

The simulation layer played a dual role:

1. Rejecting bad recommendations that would violate sweep patterns
2. Quantifying reservoir-level impact of applying ML suggestions at scale

This step turned ML from an advisory tool into an actionable decision engine that understands the physics of oil-water flow.

It also provided a mechanism for uncertainty quantification, allowing engineers to assess outcomes under varying realizations.

Transferability Across Reservoirs

While the framework developed in this study was validated using a synthetic reservoir composed entirely of horizontal wells completed with ESPs and ICDs, its core principles are broadly transferable to other upstream environments. The modularity of the data preprocessing, feature engineering, and machine learning pipelines allows for adaptation across varied completion strategies and reservoir types. For instance, the framework is well-suited to reservoirs in more heterogeneous environments. In such cases, spatial petrophysical variability can be incorporated as additional static features in both the predictive and prescriptive models.

Importantly, the emphasis on interpretability, use of reservoir-practical features, and reliance on routine operational data makes this approach particularly promising for surveillance prioritization, and reservoir simulation calibration across diverse settings.

Conclusion and Future Work

This study presents a hybrid framework that integrates machine learning (ML) with physics-based simulation to optimize water management in water flooded reservoirs. It demonstrated the feasibility and value of data-driven prediction and adaptive decision-making. The key achievements of this study include the development and validation of an ML model for predicting water breakthrough, the creation of an explainable decision tree and adaptive reinforcement learning agent for remedy recommendation, and the integration of a reservoir simulation validation loop to filter ML recommendations and quantify their spatial impact and uncertainty.

The results of this study show significant improvements over a traditional approach, with a 45-day average delay in water breakthrough per well, a 4-8% increase in cumulative oil production, a 12% increase in intervention success rate, and a 25% reduction in unnecessary actions. These gains were most prominent

in wells with high heterogeneity, dual-lateral completions, or ICD layout imbalance, conditions under which human experience alone often fails. The broader implications of this study are substantial, enabling engineers to make more informed decisions using familiar tools like decision trees, creating a scalable digital workflow to optimize rig allocation, intervention planning, and reservoir performance management, and demonstrating the value of combining domain knowledge, physics models, and machine learning to unlock value in digital transformation initiatives.

To transition this framework into field deployment, future steps include training on historical data from multiple reservoirs. Future work should also focus on integrating the reinforcement learning framework directly into the reservoir simulation model, allowing for more seamless and realistic optimization of water management strategies. Additionally, adding a systematic approach to simulation filtering of the recommendation engine's output would be crucial, enabling the identification and elimination of suboptimal or infeasible recommendations, and ensuring that only the most effective and practical interventions are proposed.

In conclusion, this work provides a practical and technically rigorous pathway to elevate water management strategies from reactive to predictive, explainable, and simulation-aligned. The demonstrated gains in oil cumulative production, water reduction, operational-efficiency and decision accuracy showcase the power of ML-physics integration in unlocking untapped potential.

Nomenclature

Table 19—Definitions of Key Terms

Term	Description
ICD	Inflow Control Device
ICV	Inflow Control Valve
ESP	Electrical Submersible Pump
ML	Machine Learning
MD	Measured Depth
Sw	Water Saturation
PI	Productivity Index
perm	Permeability
TVD	True Vertical Depth
Avg.	Average
bbl	Barrel
WC	Water Cut
SCADA	Supervisory Control and Data Acquisition Systems
PROPS.	Properties
t	Time
MDP	Markov Decision Process
SD/Std. Dev	Standard Deviation
RF	Random Forest
GBT	Gradient Boosted Trees
ROC-AUC	Receiver Operating Characteristic Area Under the Curve
SHAP	SHapley Additive exPlanations
DTC	Decision Tree Classifier

Term	Description
RL	Reinforcement Learning
Sim	Simulation

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